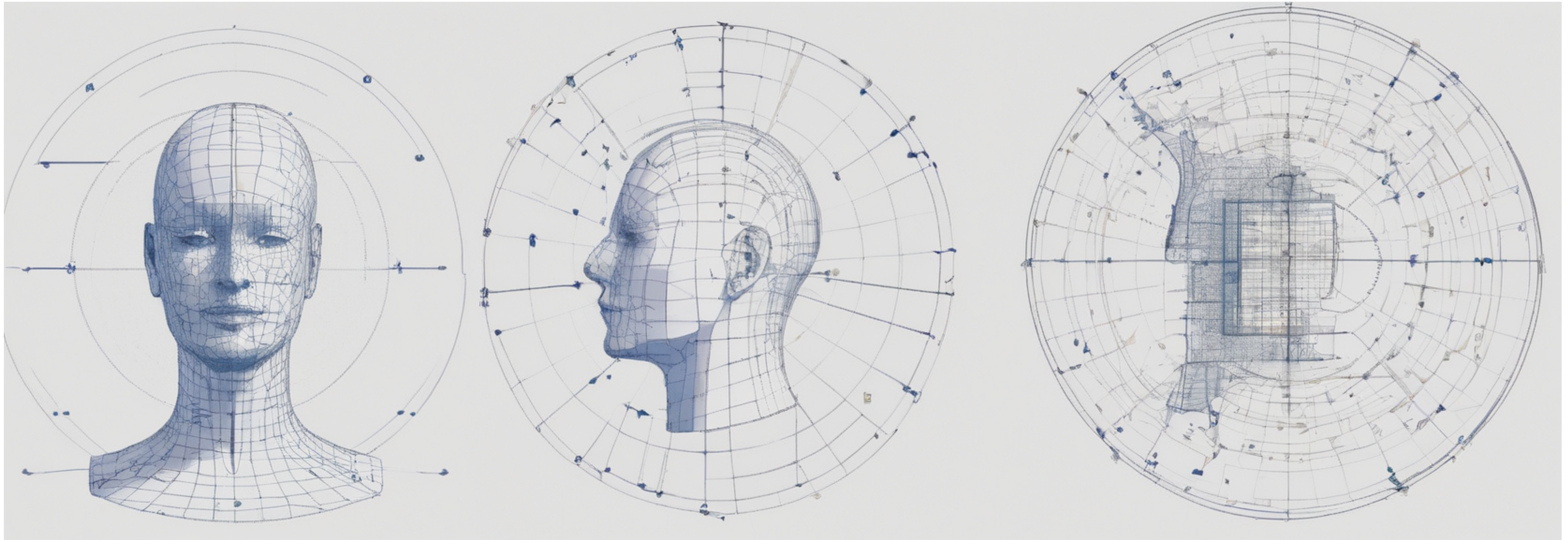




Audio Computing Lab

3D and Virtual Sound

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Overview

- Human perception of sound and space
 - ITD, IID, HRTFs, and all that
- 3D audio
 - Measuring HRTFs
 - Synthesizing 3D audio
- Virtual audio
 - Synthesizing virtual audio

What is 3D audio?

- Fooling a listener that a sound is coming from a specific location around them
- Two ways to get it:
 - Easy: Using headphones
 - Hard: Using speakers

What is virtual audio?

- Modeling the effects of being in a virtual environment
 - Includes 3D audio effects
 - Also includes room effects
 - Also includes additional environmental effects

Why bother?

- Entertainment
 - Immersive gaming, 3D movies, virtual worlds, AR/VR apps, ...
- Practical
 - Help listeners parse more audio streams simultaneously
 - Help users localize multiple sources
 - e.g. pilot discussions in place cockpits
 - For grabbing people's attention
 - E.g. in auditory display interfaces

A bit of hearing theory

- In order to synthesize 3D audio we need to know how to fool the human ear
- What are the cues that we need to use?
 - And how do we implement them?
- Multiple levels of complexity to explore ...

What is the benefit of having two ears?

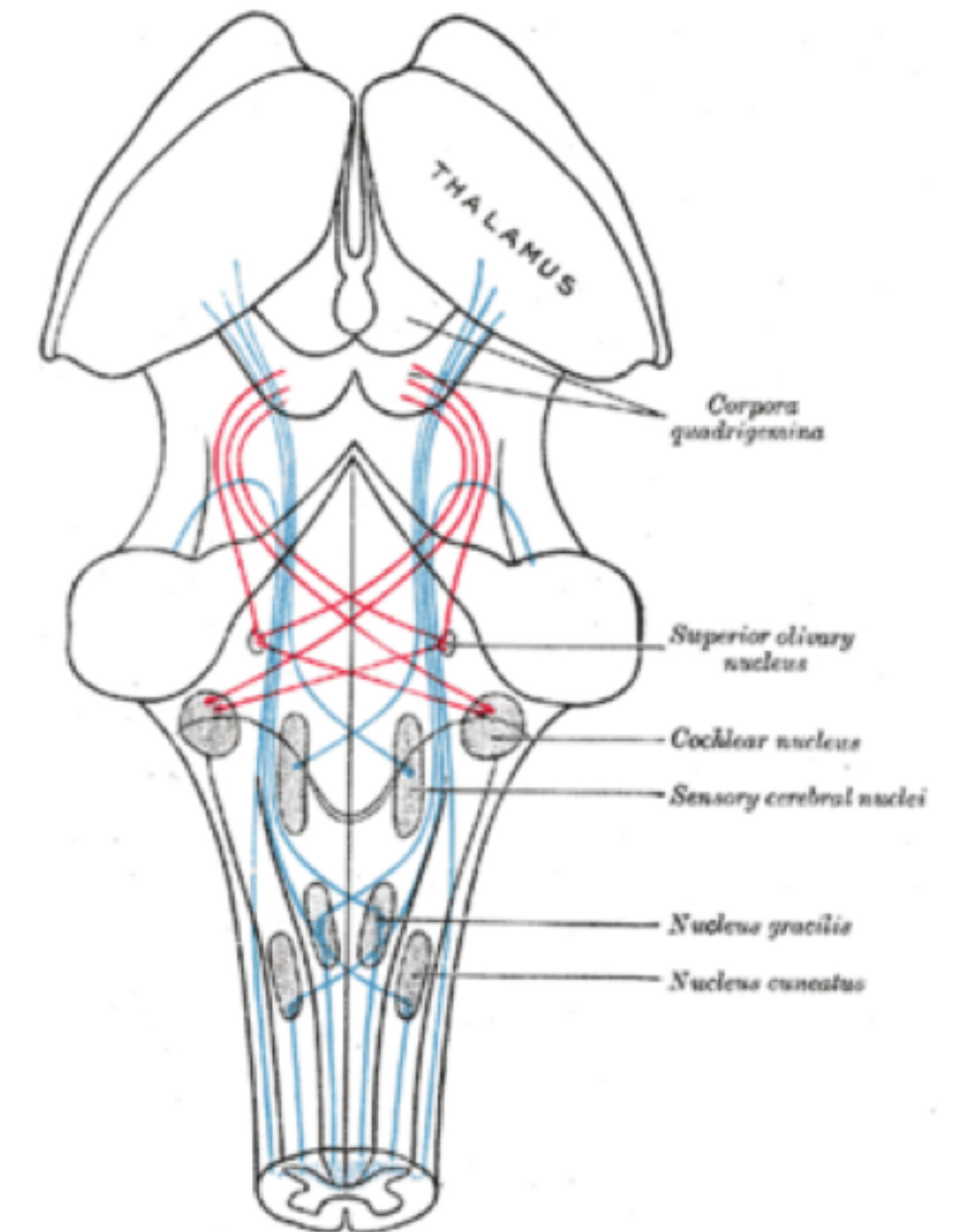


On having two ears

- Why are our ears on the sides of our head?
 - Why not one on the chin and one on the forehead?
- Horizontal placement maximizes the effect of detecting sounds over flat terrain
 - Good for left/right
 - Not so good for up/down

Fundamentally different than vision

- Unlike an eye that directly perceives spatially, our ears have to get spatial info “computed” in the brain
- Special neural circuits in the Superior Olivary Complex (SOC) compare signals from both ears

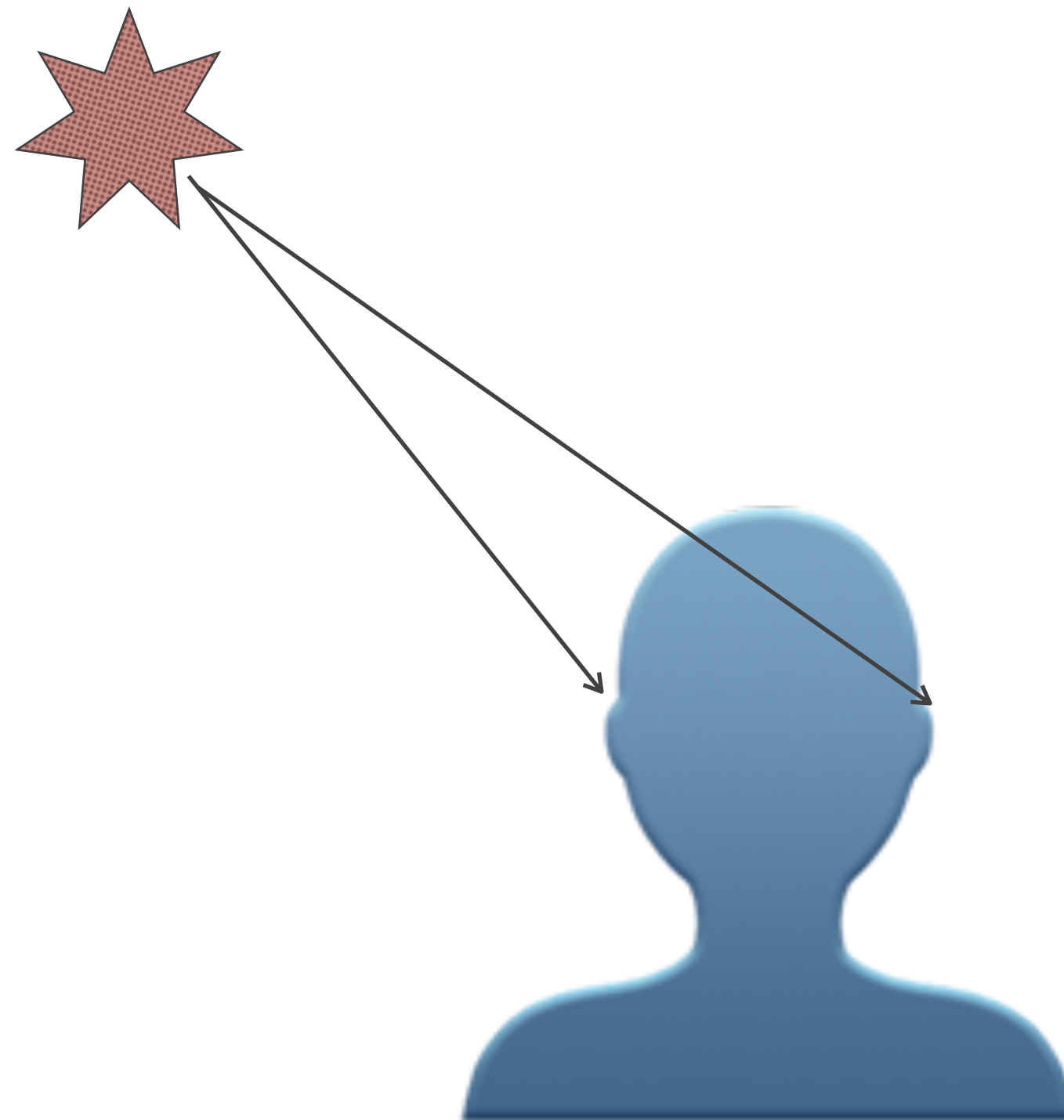


The Duplex Theory

- Formulated by Lord Raleigh (1907)
- A listener's ears receive a sound with some minor differences which act as localization cues
- The two main cues
 - Interaural Time Differences (ITD)
 - Interaural Intensity/Level Differences (IID, or ILD)

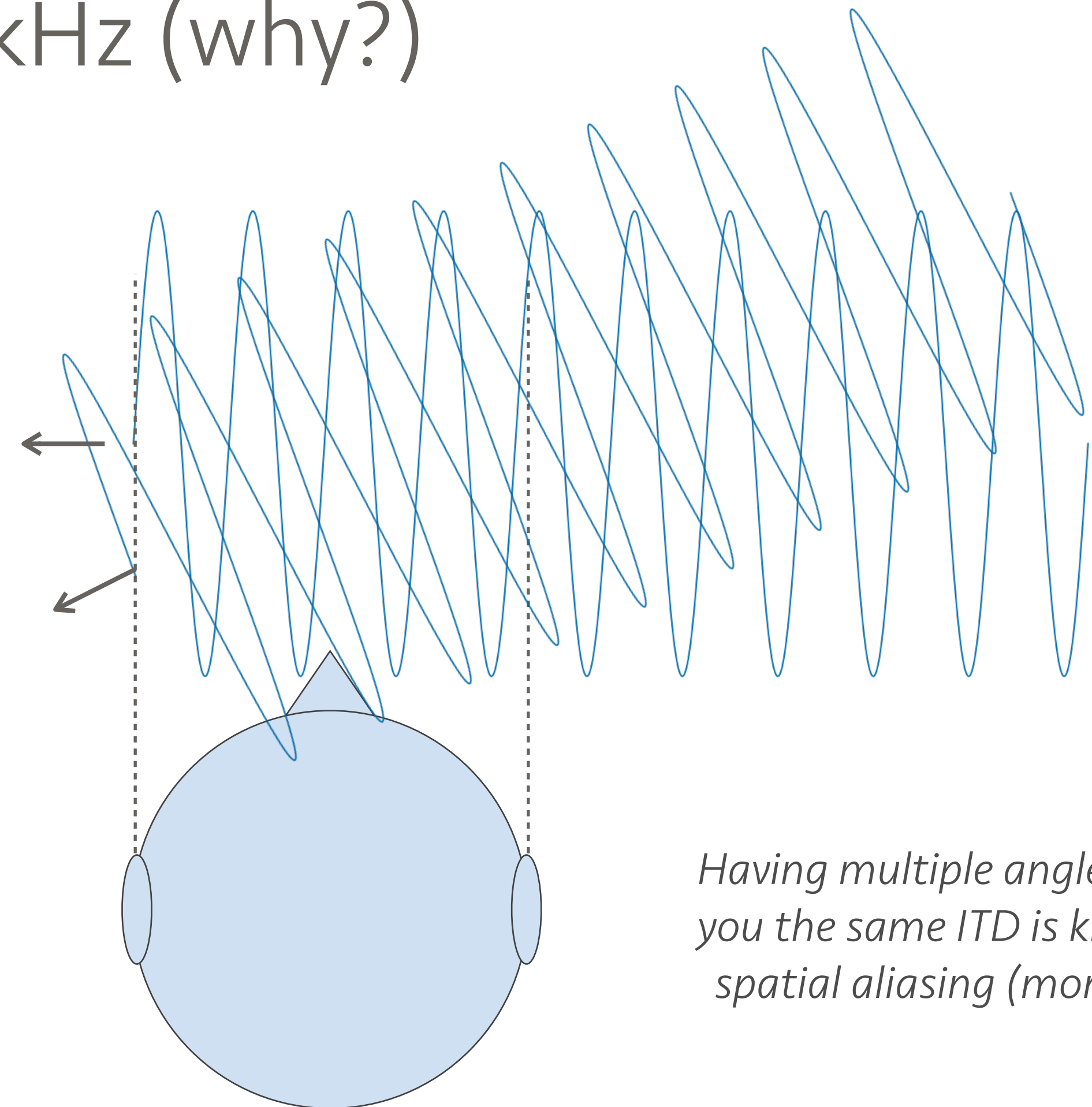
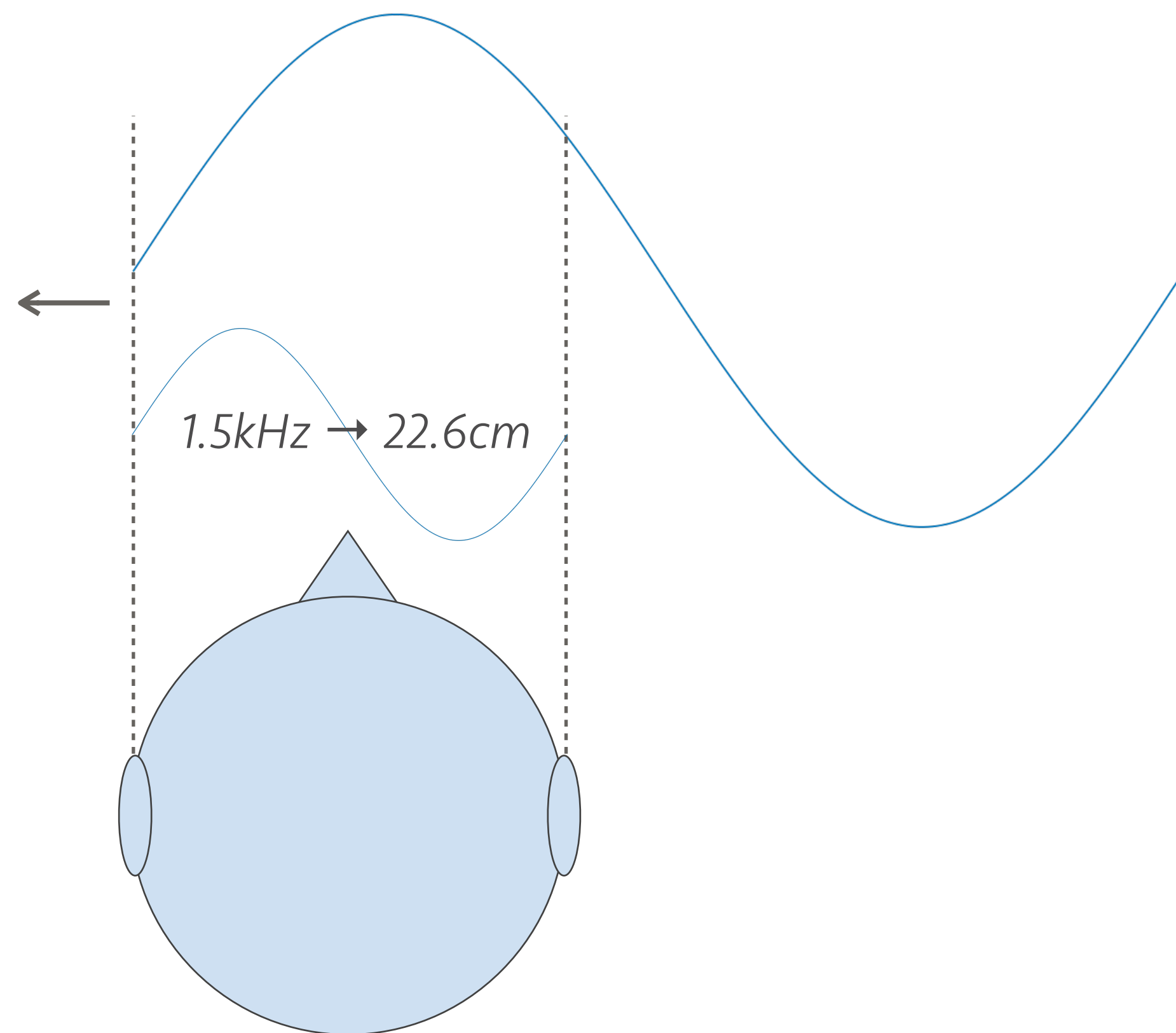
Interaural Time Differences (ITD)

- Simplest possible cue
 - Relative time difference between a sound reaching each ear



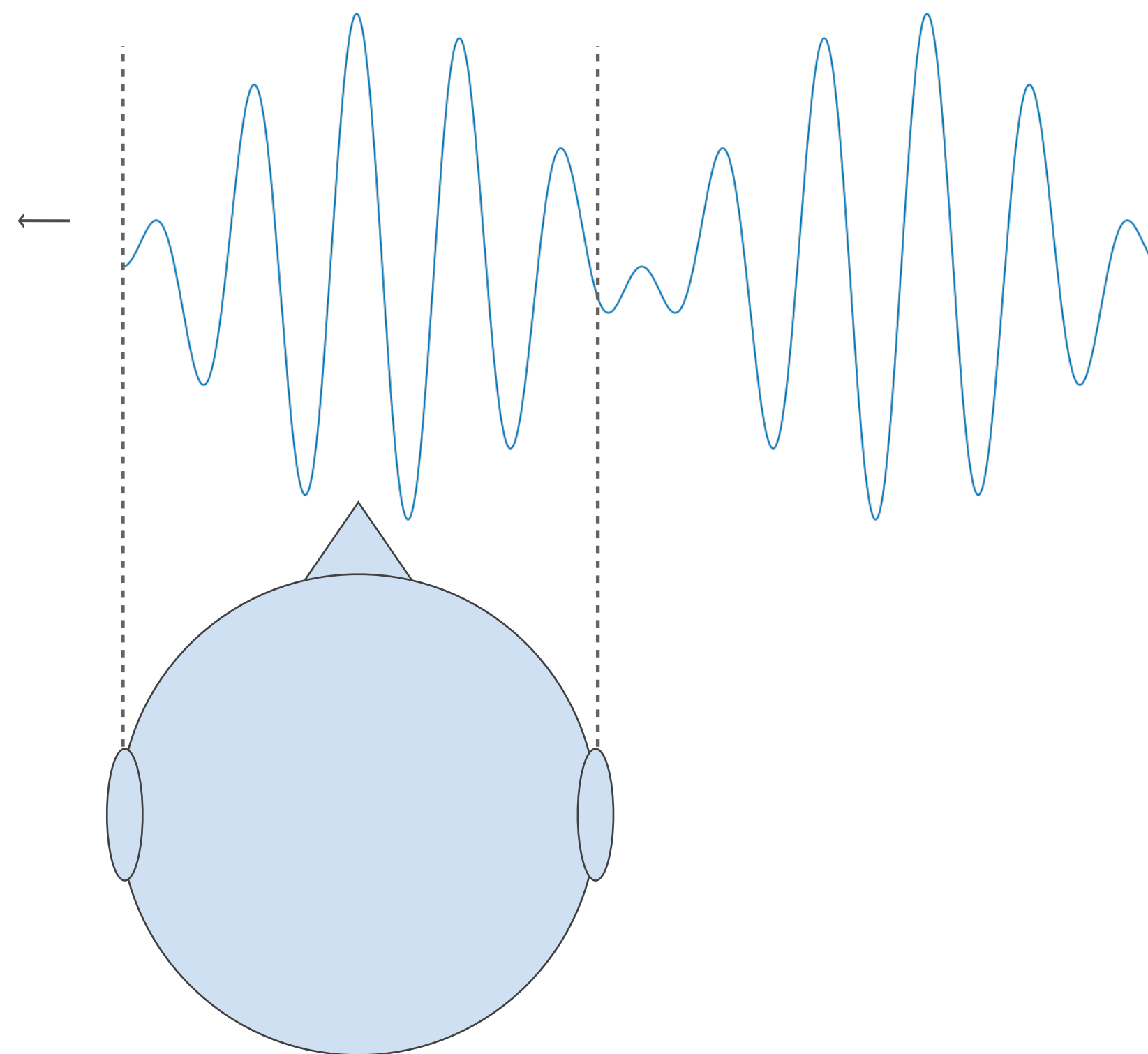
ITD as a phase difference

- Perceiving ITDs is unreliable at higher frequencies
 - Usually this starts at about 1.5kHz (why?)



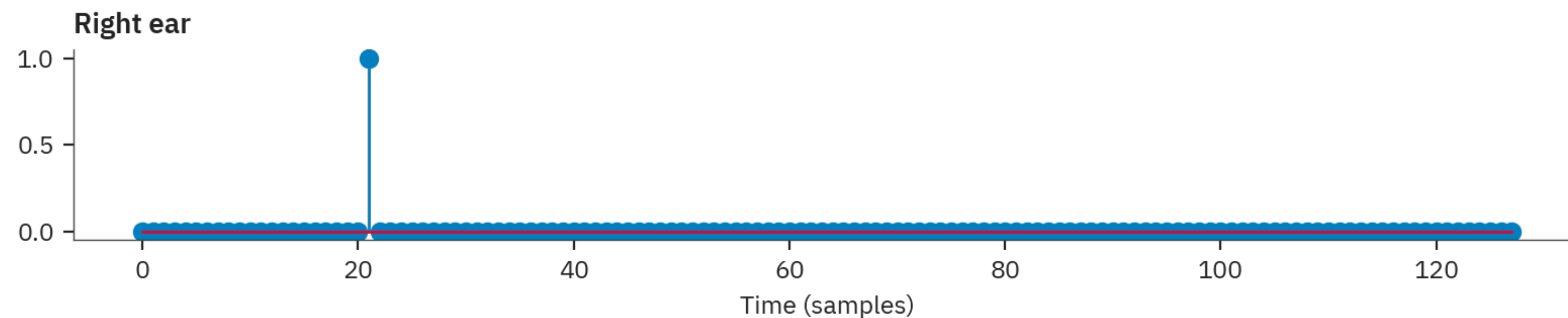
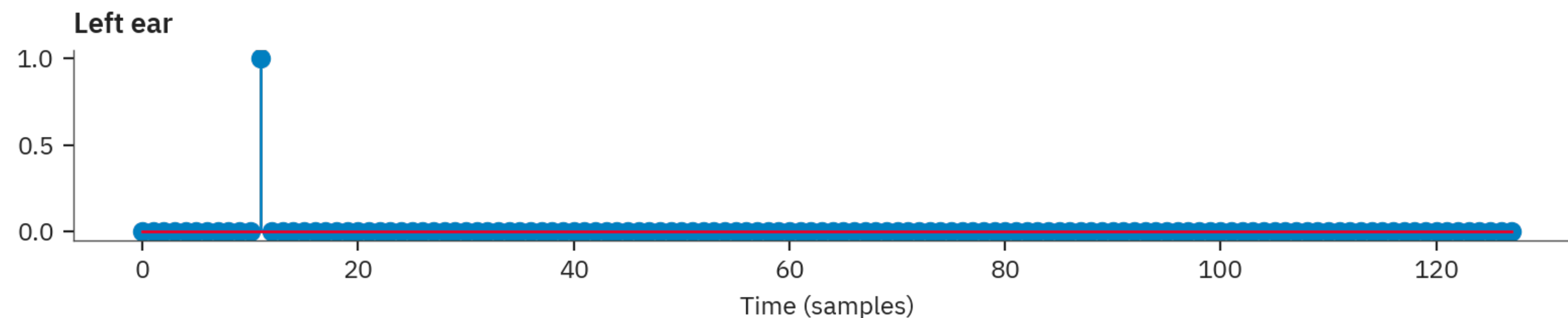
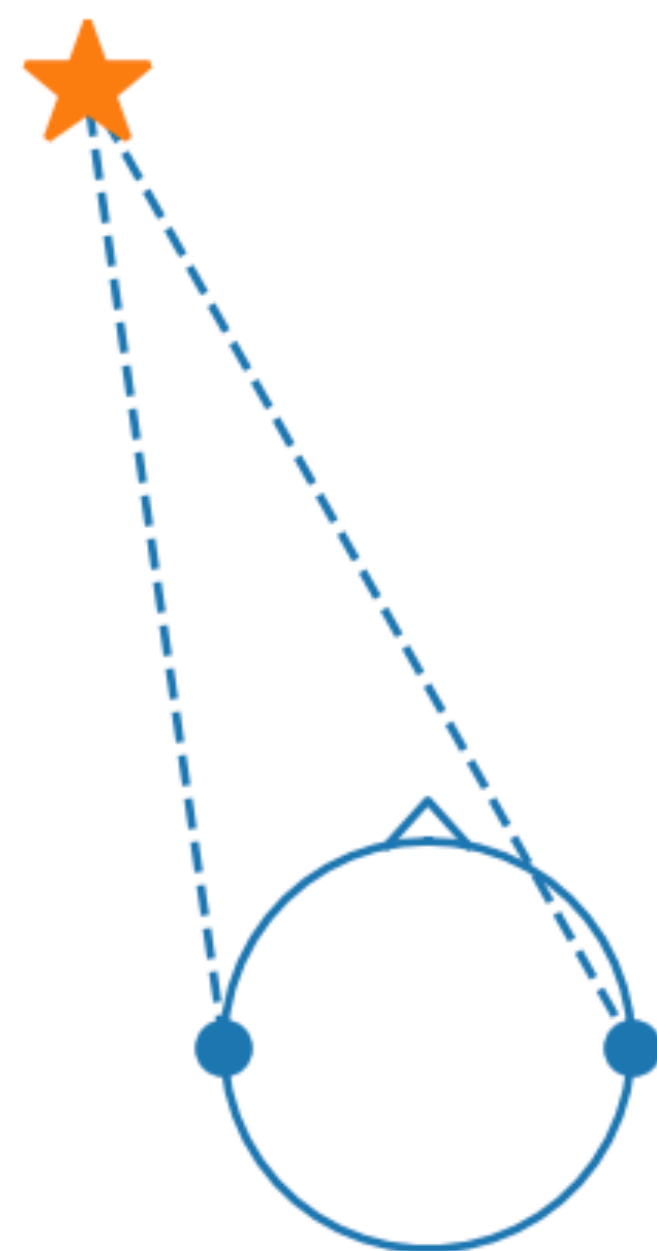
Gets resolved better in the real world

- But we don't just use phase, we track time difference with the fluctuations of sounds
 - But that's hard with some periodic sounds (sirens, beeps, etc)



How we will model it

- We can simulate ITDs with delays
 - Simple convolutions
- There will be an upper limit to the delay
 - What is it?



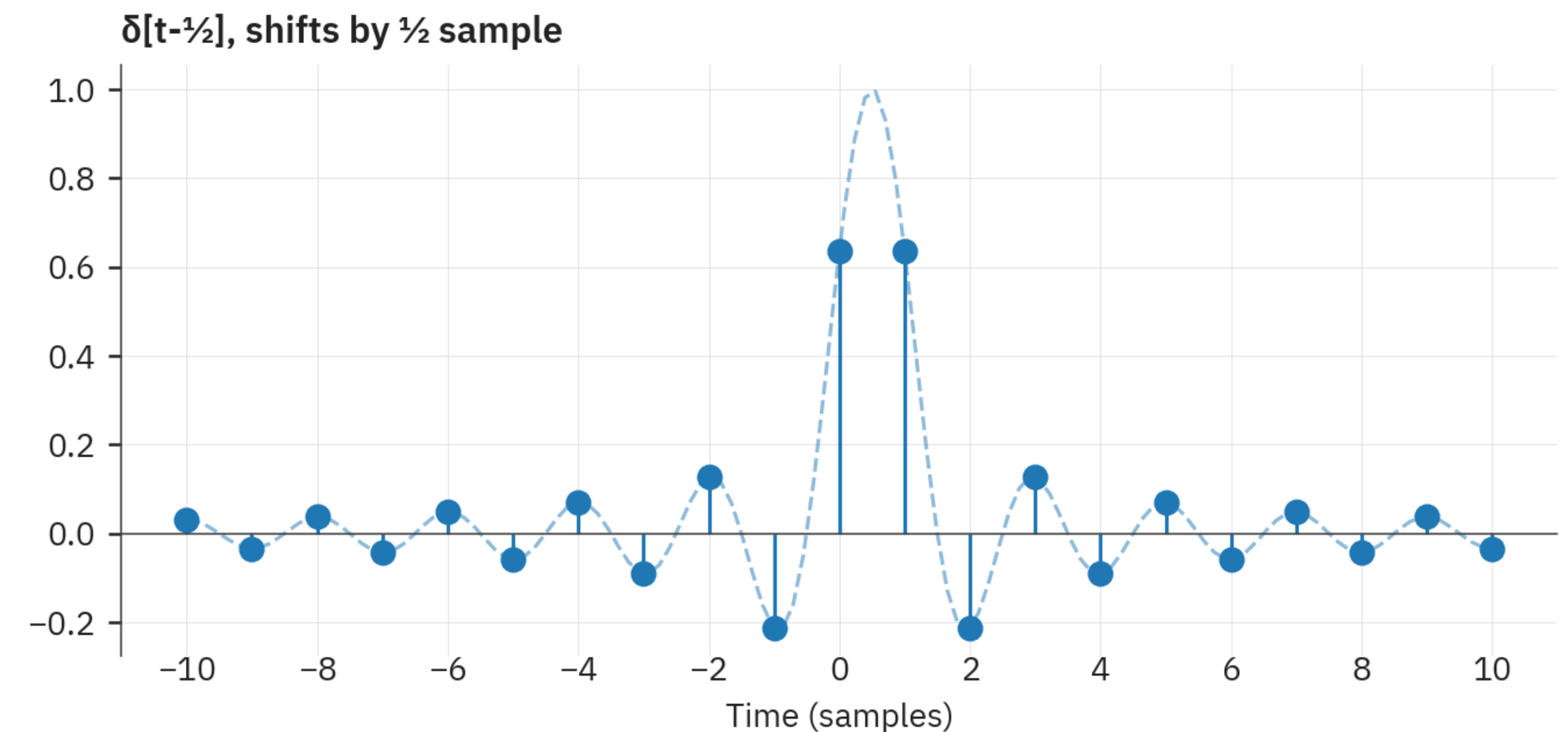
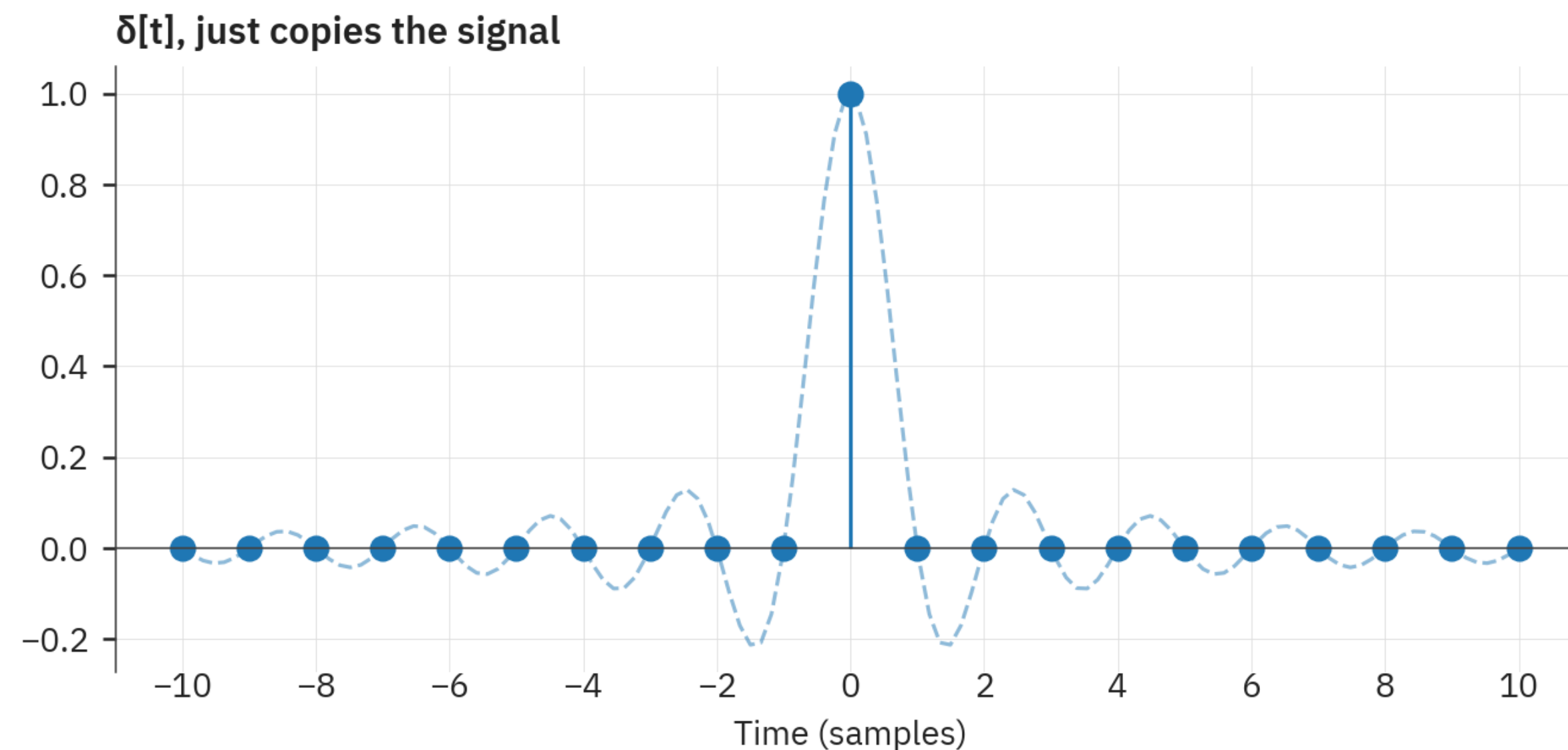
What about fractional delays?

- Time delays do not always resolve to integer samples!
 - e.g. 10ms in 44.1kHz $\rightarrow$ 220.5 samples
- How do we delay a signal by a fractional amount?
 - You might be tempted to use linear interpolation, e.g.:
$$x[t + 0.5] = \frac{1}{2}x[t] + \frac{1}{2}x[t + 1]$$
 - Why is this a bad idea?
 - This is a lowpass filter! It will change our sound!

“Proper” way to interpolate

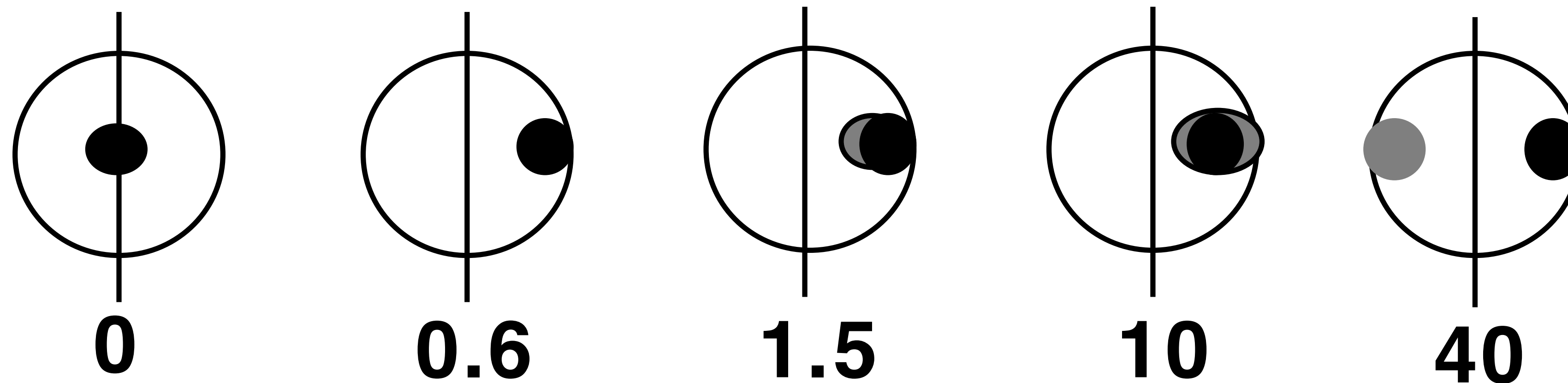
- Sinc interpolation
 - Make the lowpass filter have a cutoff of 1 and shift it

$$x[t + \varepsilon] = x[t] * \text{sinc}(t + \varepsilon)$$



Interesting perceptual tidbit

- The precedence effect (a.k.a. Haas effect)
- Up to 40msec delays register as an ITD
 - More than that and we form echo percepts



Approximate delay time to left channel (msec)

Interaural Intensity Differences (IID)

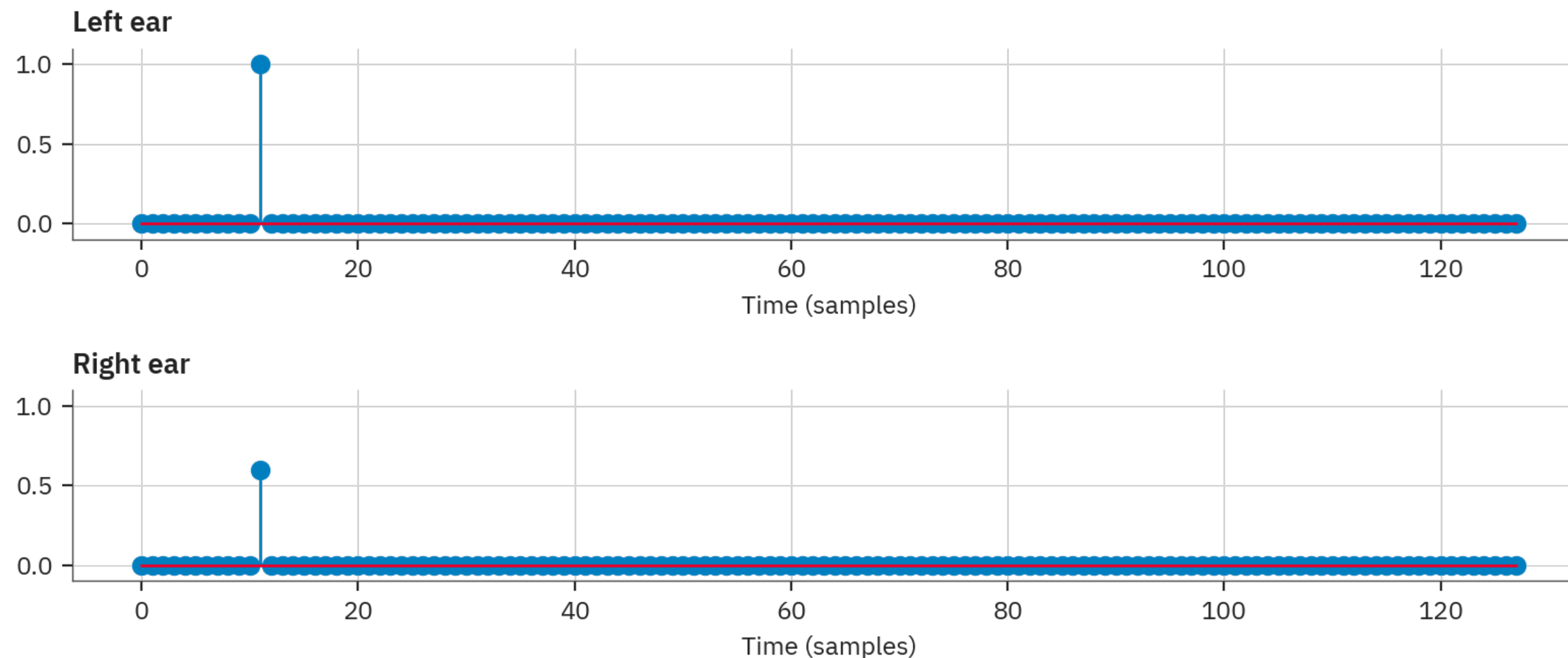
- For wavelengths smaller than the listener's head we observe sound absorption
 - High frequencies get attenuated
 - Low frequencies pass mostly unharmed
- Level differences in high frequencies are a very strong cue to help us localize sounds
 - They are called IIDs, or ILDs
 - For intensity or level

IID tradeoffs

- IIDs mostly apply to wavelengths shorter than the head of the listener
 - About a 1.5kHz cutoff
 - Lower frequencies diffract around the head
- IIDs work better when the sound source is off the *median plane* (equal ear distance)
 - Otherwise there is no relative head shadowing
 - What's an example location?

How can we model it?

- Easy to model using relative gain between ears
 - The *panpot* model
 - Ignores frequency dependencies (more later)
- Can also be implemented as left/right ear filters

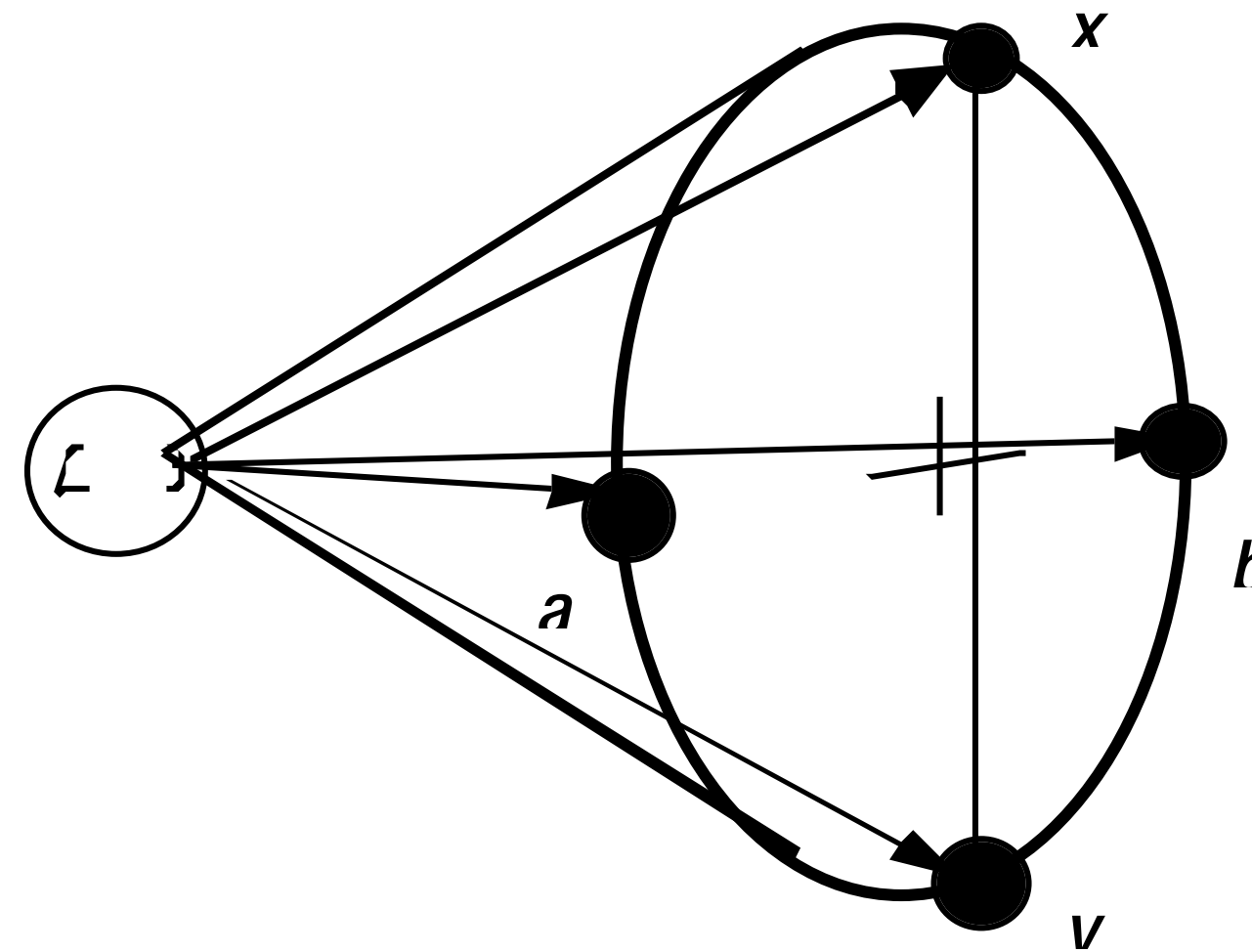


Combining ITDs and ILDs

- We can very simply combine both cues
 - This will give us a rudimentary 3D system
- Each ear gets a filter
 - Filter imposes a time delay for ITD
 - And a gain factor for the ILD

Cones of confusion

- There are parts of space that will result in the same ITD and IID values
 - We cannot distinguish sounds from these locations
 - At least not well
 - In real-life we resolve that by moving our heads



“Lateralization”

- ITDs and IIDs tend to produce *lateralization*
 - The percept of a sound on the axis between ears
 - “Inside the head” effect
- So we don’t quite get 3D sound
 - But we’re getting there!

Zoological intermission

- The Barn Owl
 - Hunts through hearing in the dark
- Can shape its face to funnel sound towards its ears
- Has asymmetrical ears
 - Which allow it to use more cues for vertical localization



Entomological intermission

- The Ormia Ochracea
 - Finds host crickets through hearing
 - Very good at localization!
- Ears are 0.5mm close
 - How does it use ITD/IID?
- Coupled eardrums create new cues
 - Currently used as model for new mics

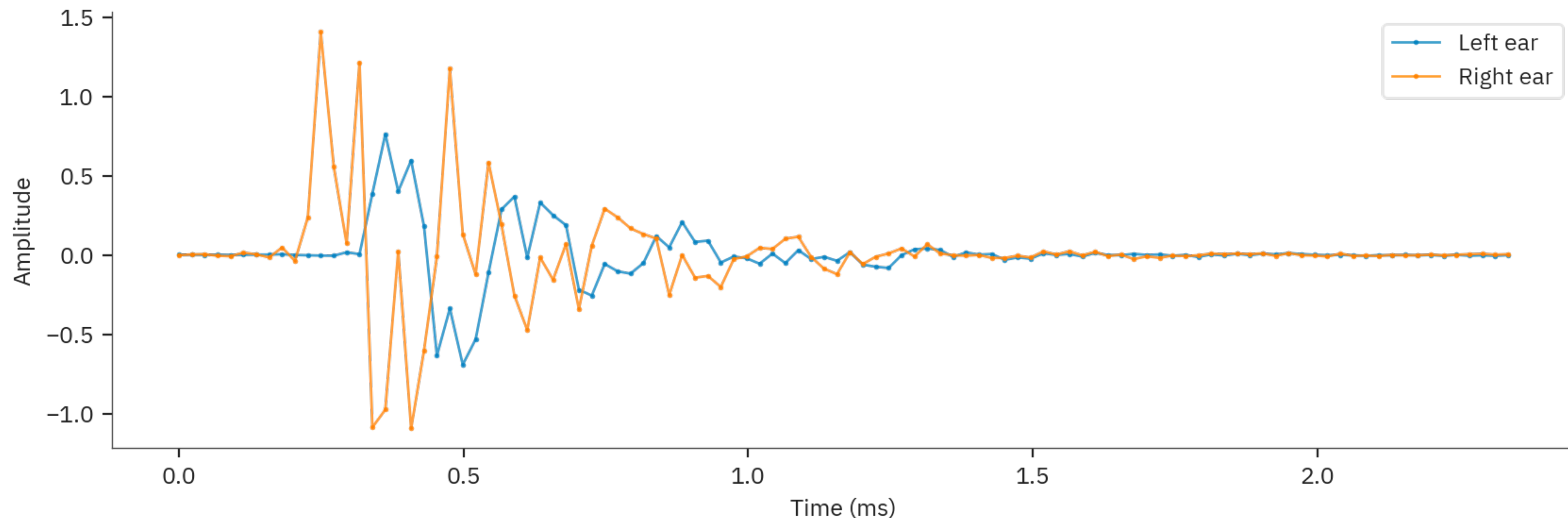


One cue to rule them all!

- ITDs and ILDs can be insufficient
 - Lateralization issues
 - Very simple model of environment
 - Our ears adapt to localize and are in fact a lot smarter
- Head Related Transfer Functions (HRTFs)
 - Incorporating more, and finer cues for localization

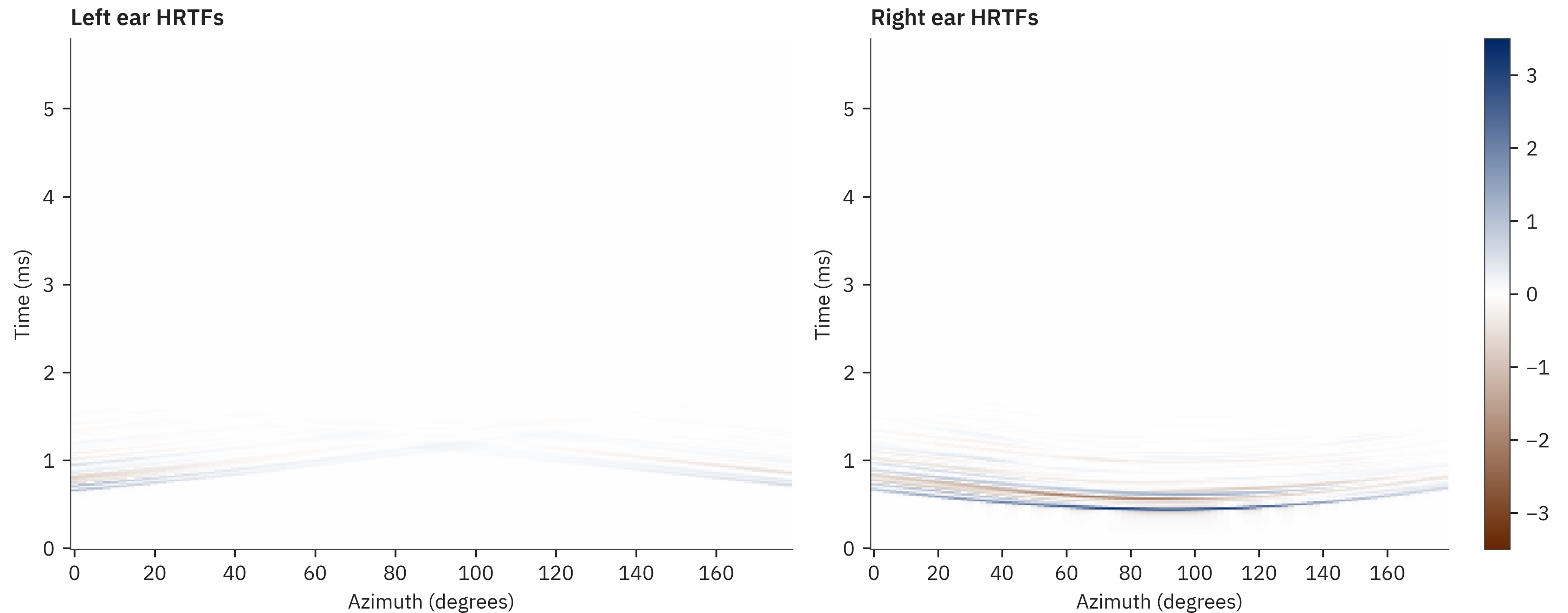
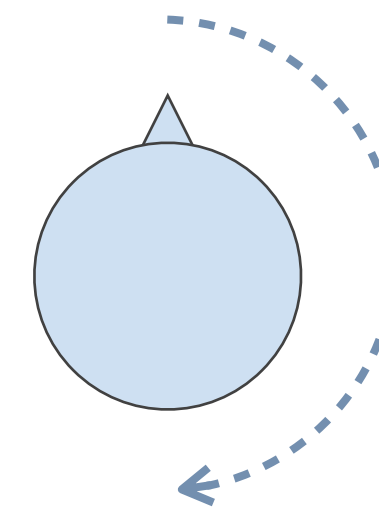
What do HRTFs capture?

- Many effects relating to our body shape
 - Funneling by the ears, reflections off our shoulders, sound absorption from head, effects from hair, ...
 - They also incorporate ITDs and ILDs



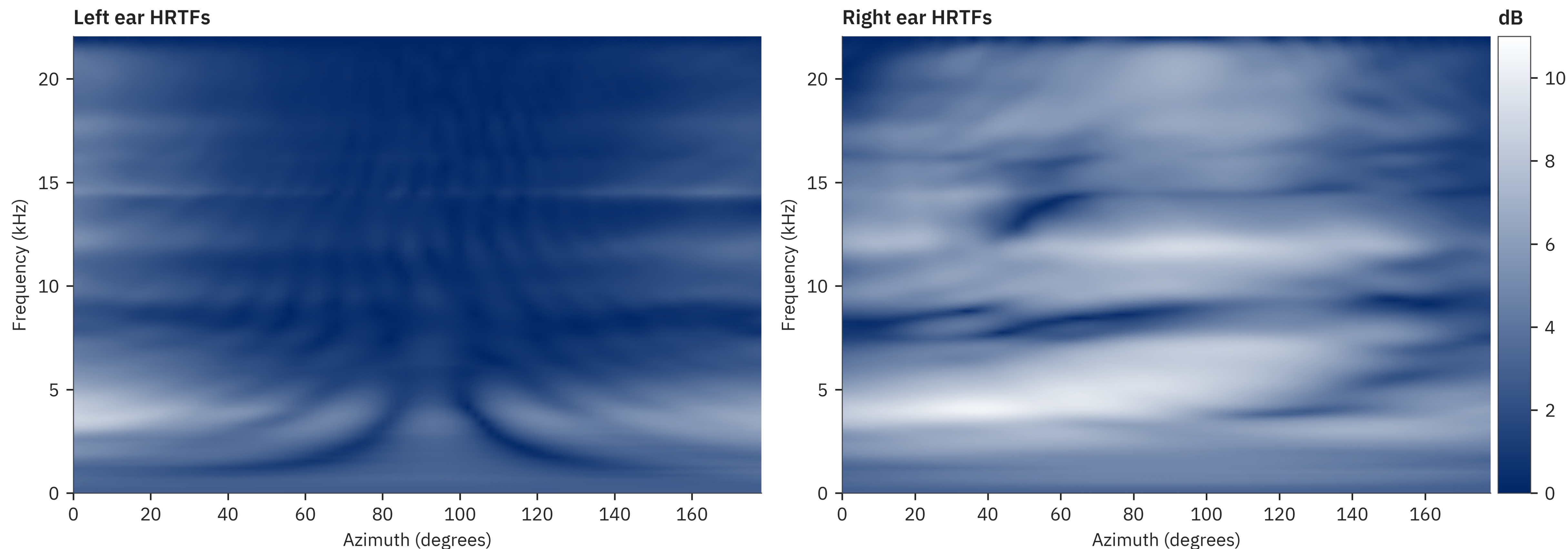
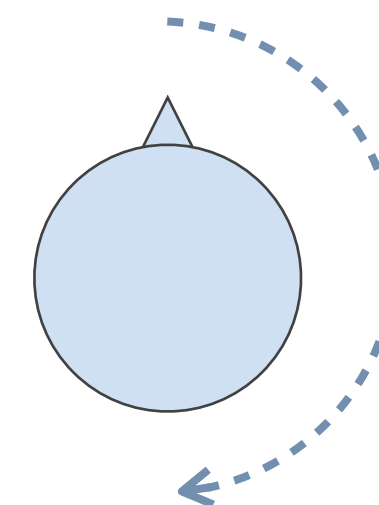
How do they look like?

- Sweep from front to back (right side)



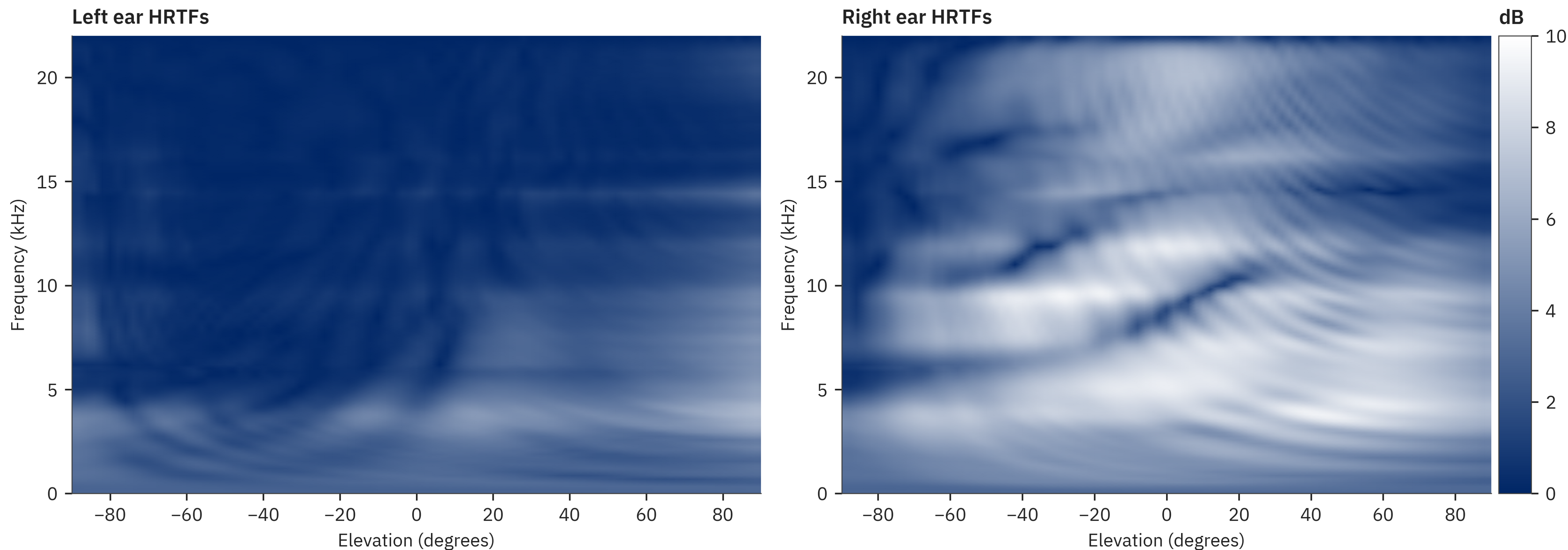
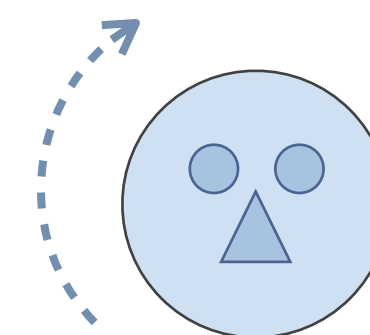
How do they look like?

- Sweep from front to back (right side)



How do they look like?

- Sweep from down to up on the right



Can we generalize?

- Each person has a different head/torso shape
 - We often just use average HRTFs
 - They won't work for everyone
 - Being average helps in this case!
- But how do we apply HRTFs?

Solution 1: Binaural recordings

- Use a dummy head to make 3D recordings
- Or stick microphones in your ears
 - (but, in general, please don't stick stuff in your ears!!)



Solution 2: Measure and apply HRTFs

- If we measure real HRTFs we can then use them on arbitrary sounds to make 3D audio
 - Just apply them as filters to generate left/right/signals
- Two ways to measure HRTFs
 - Measure a dummy head's HRTF
 - Will give you an “average” HRTF set
 - Measure your own HRTFs
 - You then have a personalized copy



How do we measure HRTFs?

- Same process as measuring room responses
 - Setup microphones in dummy or human subject
 - Play MLS/chirp/etc from different locations
 - For each location measure the response
 - You should remove the speaker/mic responses though
- Pro tip
 - Do that in an anechoic chamber
 - Why?



In math

- We record:

$$y_{\theta,\varphi}[t] = h_{\theta,\varphi}[\omega] * x[t] \quad \leftarrow \text{MLS, chirp, etc.}$$

$$Y_{\theta,\varphi}[\omega] = H_{\theta,\varphi}[\omega] \cdot X[\omega]$$

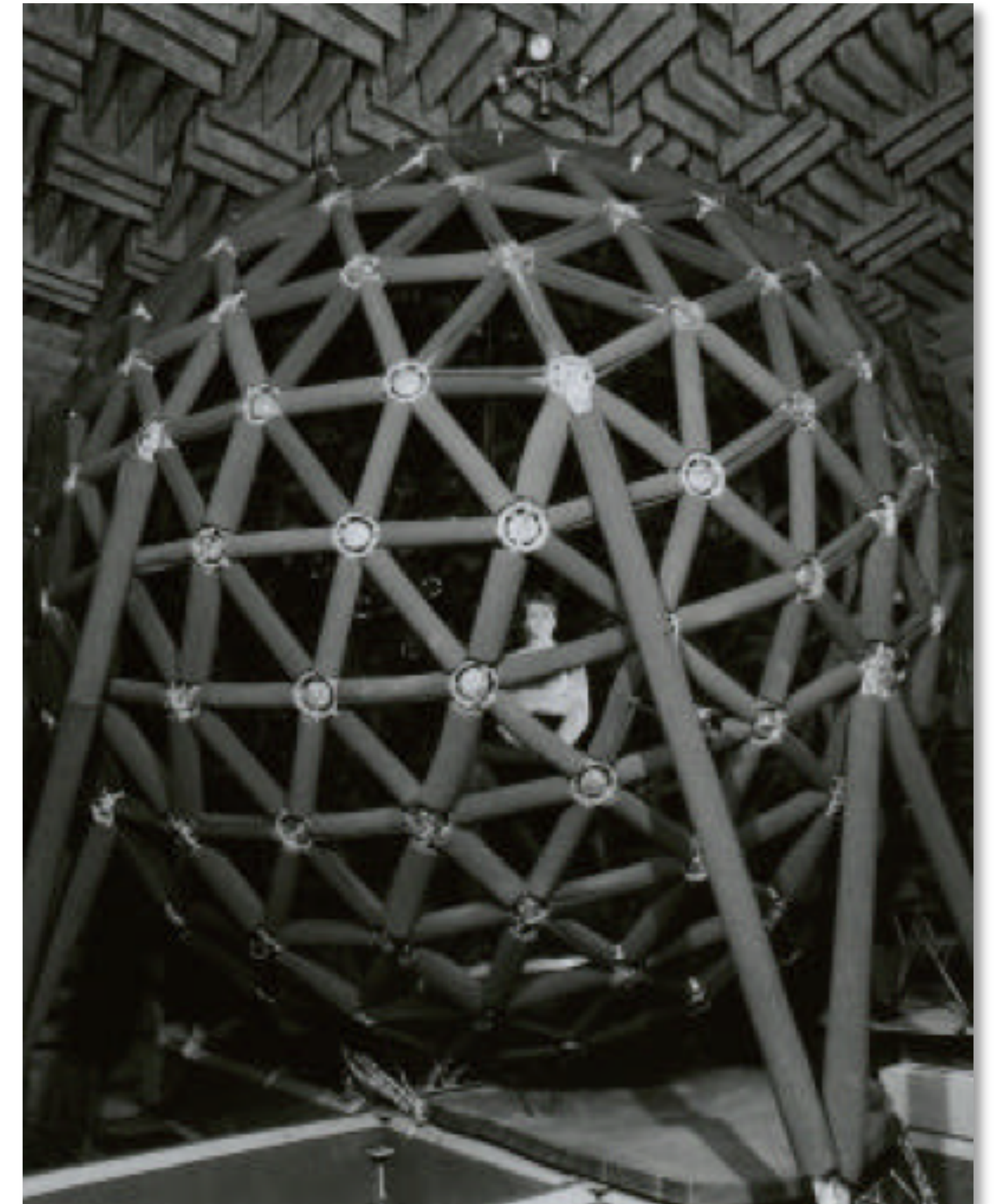
- We deconvolve with:

$$H_{\theta,\varphi}[\omega] = X^*[\omega] \cdot Y_{\theta,\varphi}[\omega]$$

- We must remove speaker/mic responses
 - Use inverse filters of these responses
 - How do we measure these?

One complication

- This requires some serious lab space



One more complication

- We measure the transfer function from the sound source location all the way to the ear
 - But none of the room effects!
- HRTFs only work with headphones/earphones
 - They do not compensate for effects from distant loudspeakers
 - We need to find ways to incorporate that
 - More later

Synthesizing 3D audio

- Pick a location to position a source
 - Usually azimuth (left-right) & elevation (up-down)
- Apply appropriate filters from HRTF set
 - Space saving hack: there is left/right symmetry so there is no need to keep all of the HRTFs, just keep one side
- What about moving sounds?
 - We need to change the filters dynamically! How?

The *Fast Convolution* algorithm

- Allows us to:
 - a) Perform complicated convolutions really fast
 - b) Implement convolutions with dynamically-changing filters
- Core algorithm in many audio processes
 - e.g. reverb algorithms, echo cancellers, etc.

Convolution in the frequency domain

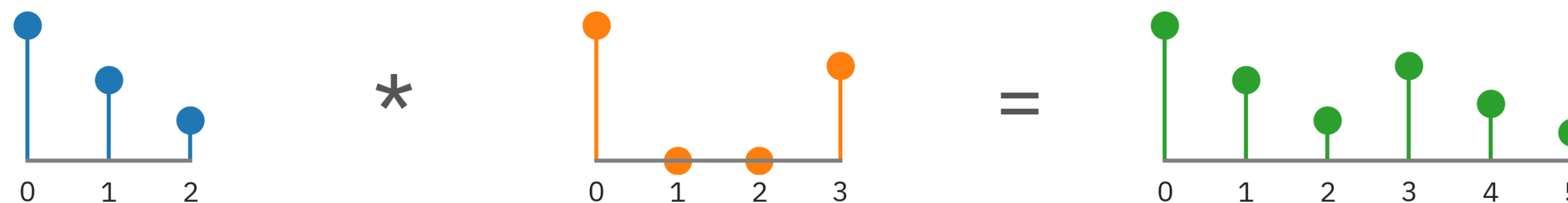
- Convolution can be sped up significantly using the FFT
 - Perform convolution in the frequency domain
 - Computational complexity drops significantly

$$z = x * y \Leftrightarrow \text{DFT}(z) = \text{DFT}(x) \cdot \text{DFT}(y)$$

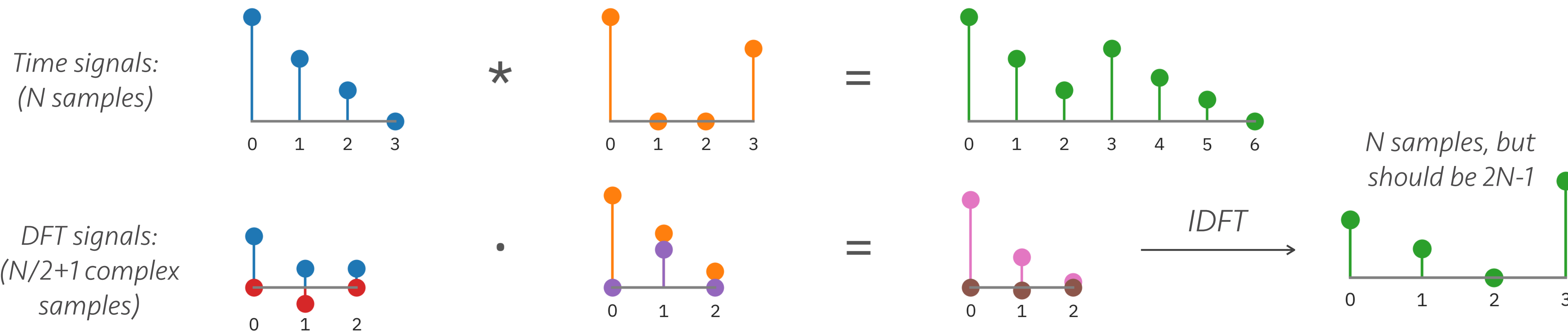
- But is this useful for our case?
 - No, results in very large FFTs, doesn't allow for changing filters
- Also, this performs *circular convolution*
 - Which is a slightly different operation

The details of DFT convolution

- Convolution output has sum of lengths minus 1

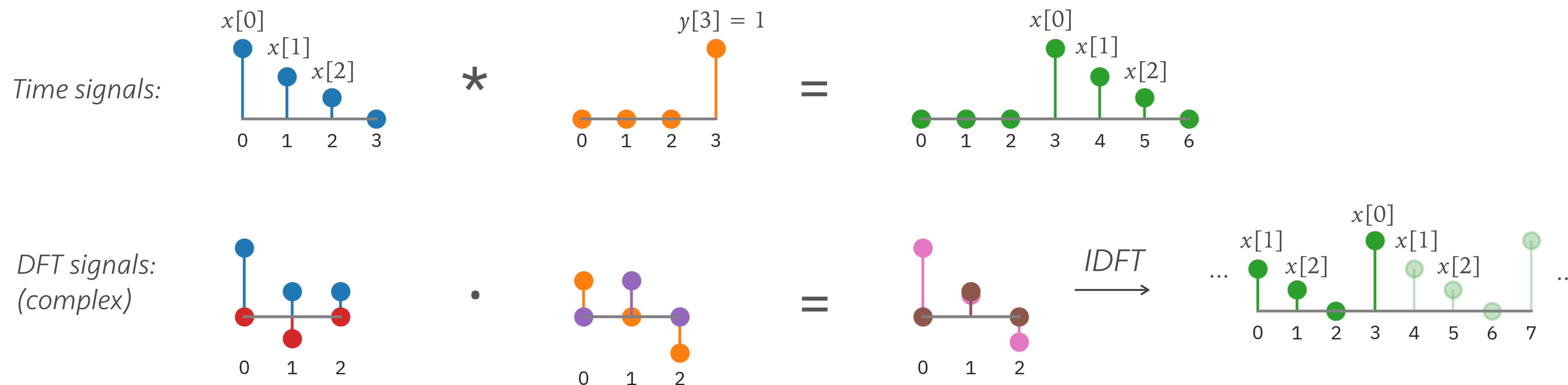


- But if we use a DFT, the output will be shorter!
 - Also to use the DFT we need to zero pad x to be as long as y



Circular convolution

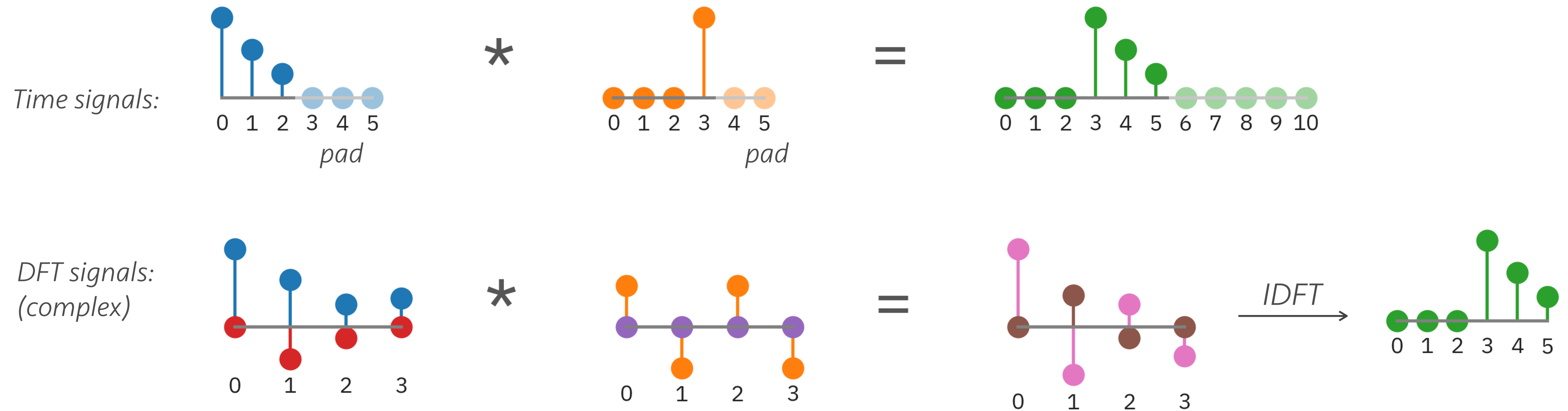
- Remember the DFT is periodic
 - The tail of the convolution is “wrapped around”



- This is also known as *time aliasing*

The solution: Zero pad

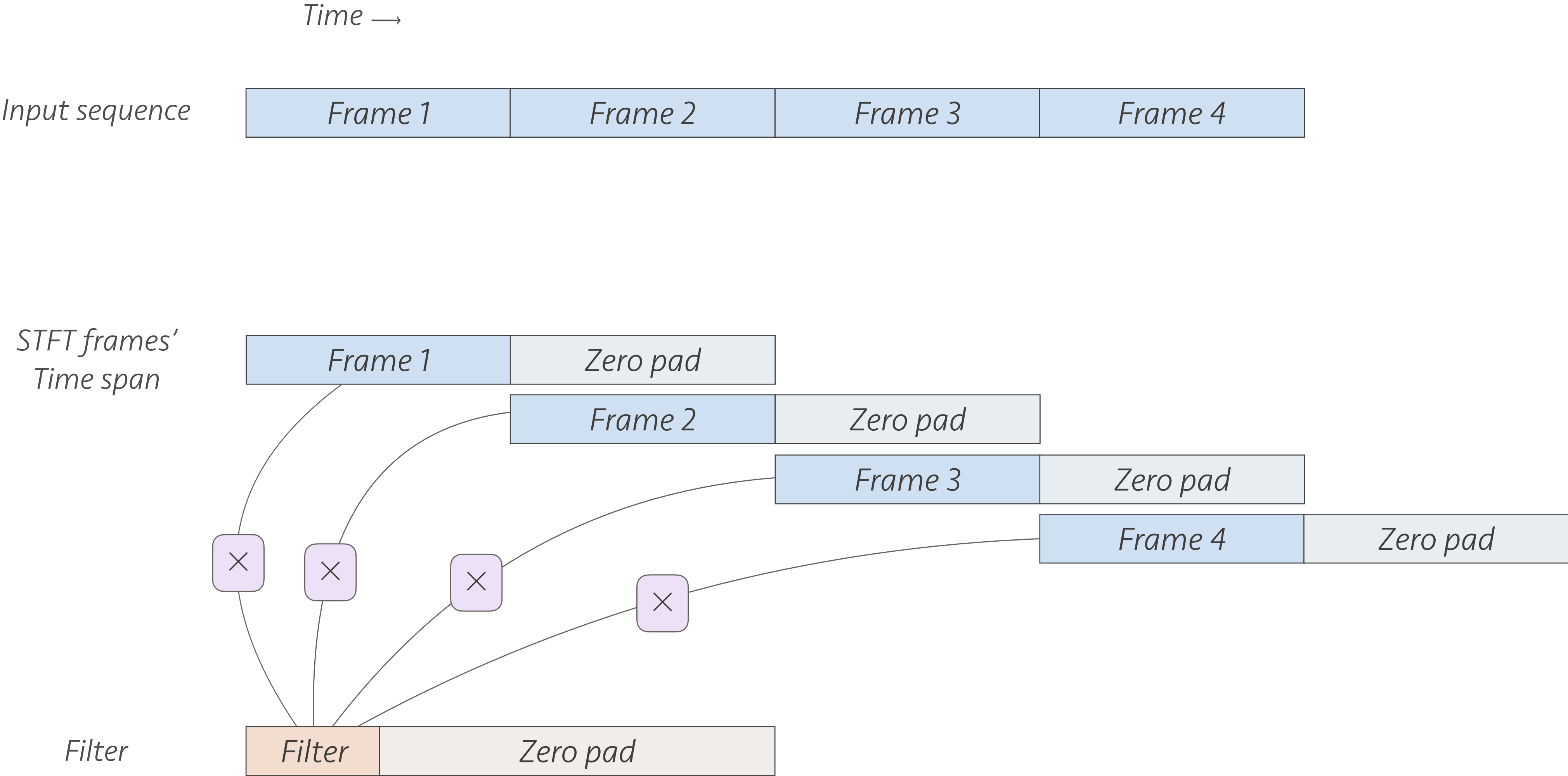
- Adding zeros to our inputs won't change the result
 - But it will take care of the length issues



The “fast convolution” algorithm

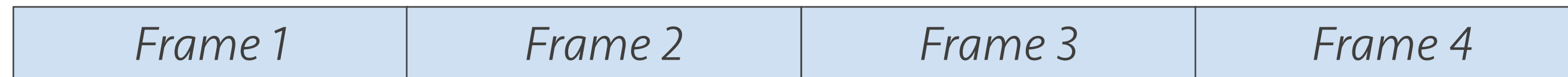
- Step 1: Perform a STFT
 - No window, hop = frame size = filter size = zero pad amount
- Step 2: Convolve in the frequency domain
 - Element-wise multiply each STFT frame with filter's DFT
- Step 3: Overlap-add to get output waveform
 - No synthesis window
 - (otherwise the convolution identity doesn't hold)

Picture aid

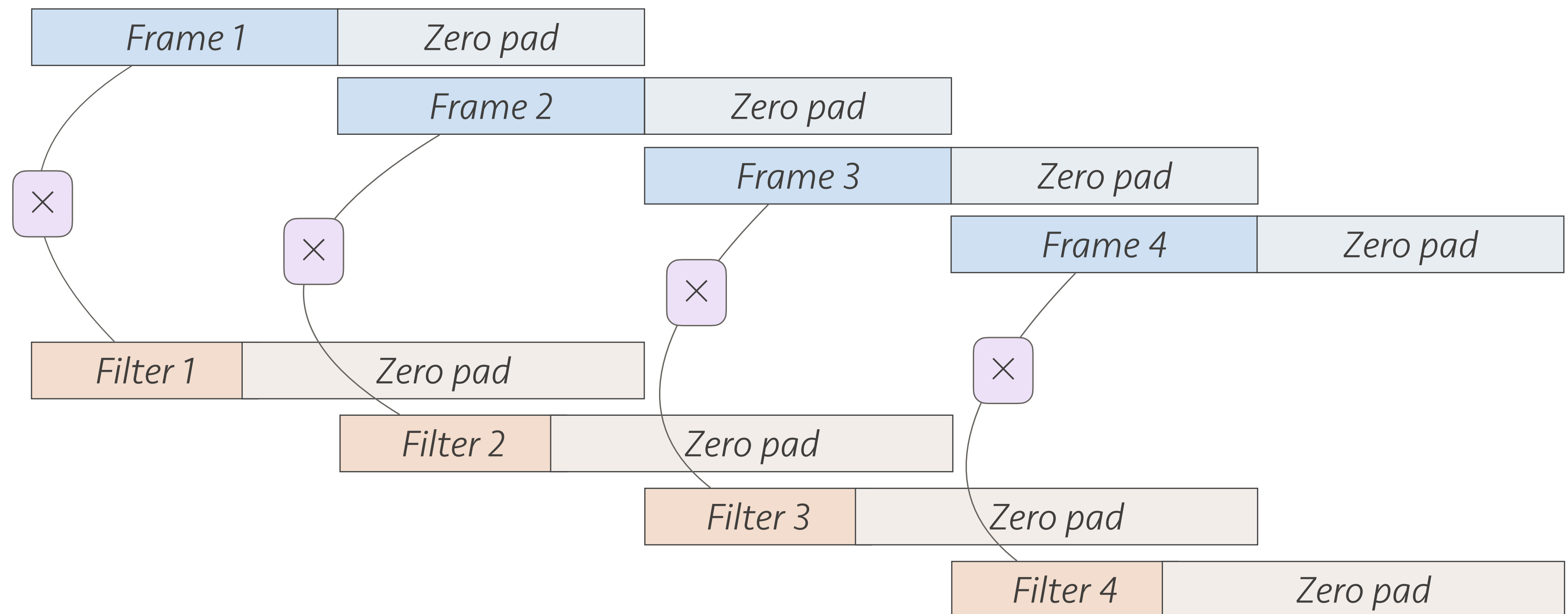


With dynamically changing filters

Input sequence



STFT frames'
Time span



Filters

Common problems in 3D audio

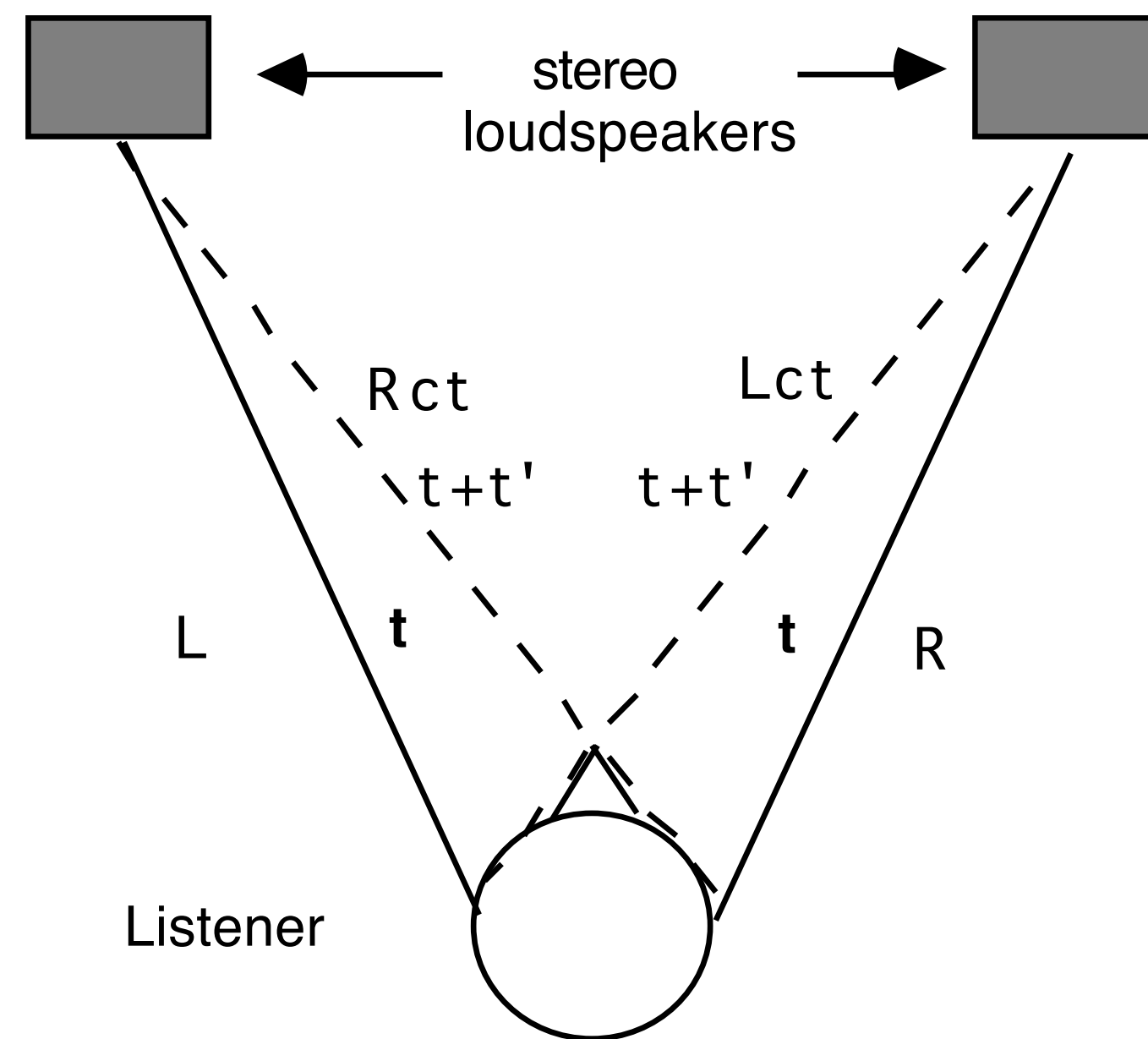
- Response mismatch
 - People with funny head shapes
 - Poor reproduction (e.g. bad headphones, MP3s)
- Front/back confusion
 - Really prominent for many people
- Head movements
 - Change the relative angle of a source

Compensating for head movement

- We can track the listener's head movements
 - Using a simple sensor on the headphones
 - Or using computer vision to measure head pose
- This allows us to find the angle between the virtual source and the rotated use head
 - One drawback: Time lag
 - One advantage: We can resolve localization ambiguities
 - We use head movements to deal with ambiguities

What about speaker 3D audio?

- We need to perform *crosstalk cancellation*
 - Use “negative” signals to get the HRTF filtering at the ears



- What are the complications here?

Complications with speaker systems

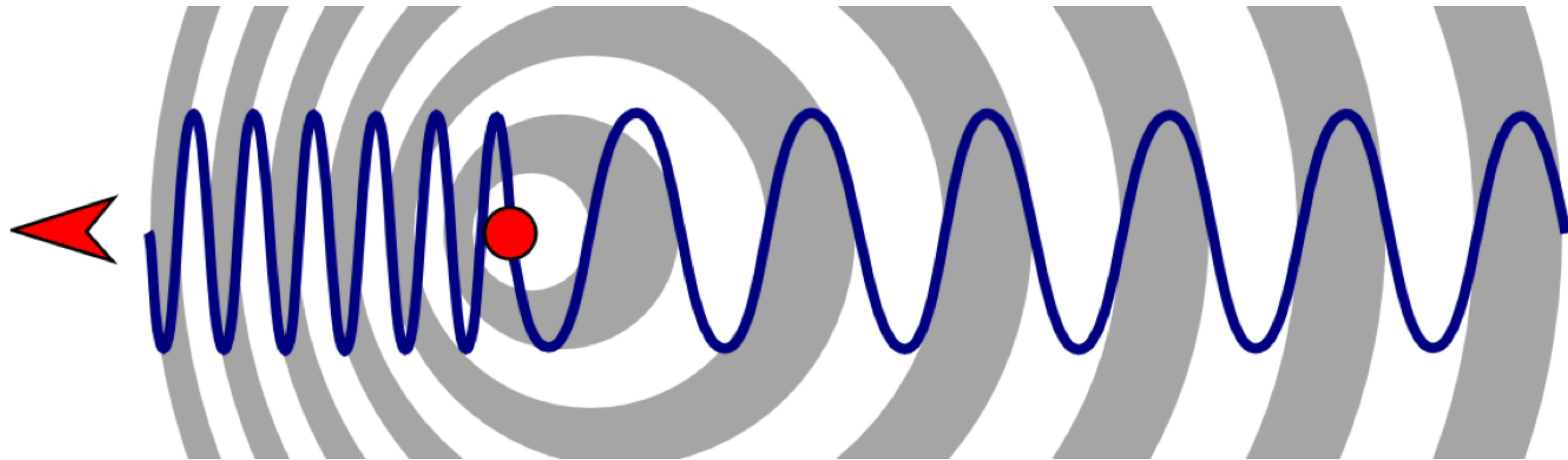
- Head movements
 - We need to compensate for moving ears!
 - Not trivial to cater to multiple people simultaneously
 - E.g. you won't get 3D sound in a movie theater
- Room effects
 - Speaker output gets convolved with room
 - and speakers ...
 - Difficult to compensate for all that

Moving towards virtual sound

- 3D sound models source-to-ear effects
 - Created 3D percept, but this is not the whole story
- There are more cues that we use to localize
 - Movement cues, distance cues, context cues, ...
- Proper virtual audio also models these cues

Movement cues

- Moving sources exhibit an additional important cue for localization
 - The Doppler effect



Distance cues

- We can also perceive how far a sound is
- Static cues
 - Level, amount of reverberation
- Dynamic cues
 - Change of source angle by head translation

And some more context cues

- Room acoustics
 - Sounds in different parts of a room sound different
- We can use HRTF filter on all the reflections
 - Overkill, but makes a difference
 - And you know how to do that now! :)

Virtual sound can be complicated

- Lots effects that combine
 - Not completely clear which are necessary
 - Depends on usage scenario
 - Also not fully clear how they all interact
- Still an open problem
 - But sounds pretty good as is

Surround sound

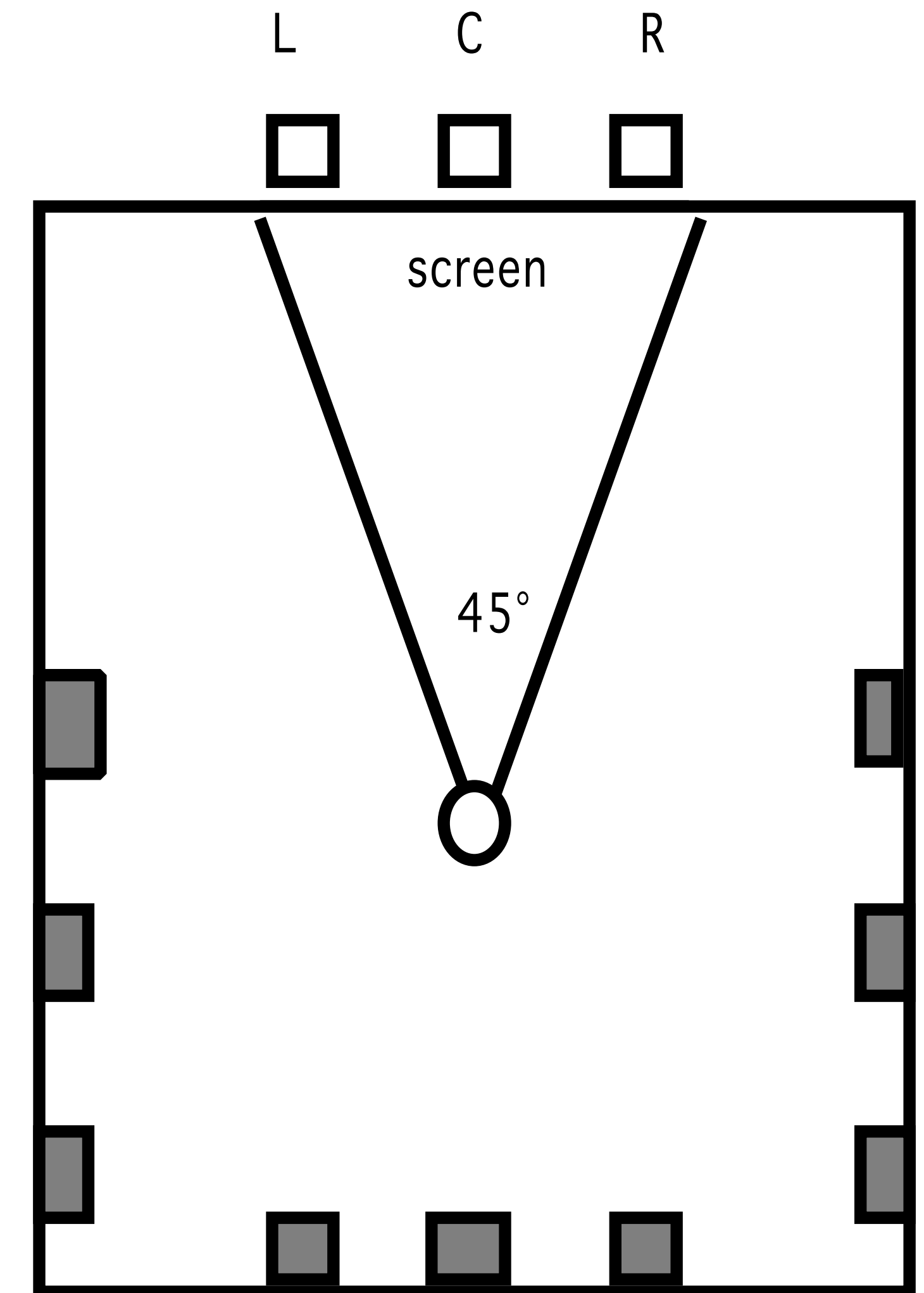
- Potentially simpler approach
 - Localization takes place using multiple speakers
- Common setups
 - 5.1 / 7.1 sets
 - Stereo “surround”
 - Avoid like the plague!
 - Ruins stereo imaging



A virtual acoustic room setup

Theater surround sound

- Front channel for dialog
 - Ensures consistent localization
- Side and rear channels for FX
 - Also ambience sounds
- One of Dolby's claims to fame



Recap

- Some of the basics of 3D perception
- HRTFs
 - How to measure them
 - How to use them
- Additional ties for virtual audio
- Surround sound

Reading material

- 3D Sound for Virtual Reality and Multimedia
 - http://human-factors.arc.nasa.gov/publications/Begault_2000_3d_Sound_Multimedia.pdf

Next lab

- Let's make some 3D sounds!
 - Remember to use your headphones/earphones
 - You won't be able to hear the results otherwise